Akoizo Original Practice - Form 3 - Answer key

Division C | Wright Stuff | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) For a fixed motor mass, what does a longer, thinner motor provide?

A. More turns and lower torque

B. Fewer turns and higher torque

C. More turns and higher torque

D. No change

Correct: A

2. (2 points) A team forgets to declare a flight as a trim flight. How is it treated?

A. As an official flight

B. As a trim flight

C. It is not timed

D. It is repeated

Correct: A

3. (2 points) An airframe weighs 7.8 g. What must happen?

A. Remove the motor

B. Use a heavier motor

C. Add at least 0.2 g so it meets the 8.0 g minimum

D. Nothing

Correct: C

4. (2 points) A plane sinks during cruise even with turns left. What propeller change may help?

A. Add weight to the blades

B. Remove the bearing

C. Reduce diameter or pitch so cruise torque can turn it

D. Increase pitch further

Correct: C

5. (4 points) Explain why a slower plane can stay up longer on the same motor.

Unit: Wings and slow flight.
Worked-case data: A 9.5 g plane (with motor) has 0.040 m^2 of wing and C_L = 0.8. Find its level-flight speed (ρ = 1.2 kg/m^3).
Separate investigation observations:
Plane 1: Wing area 0.030 m^2; cruise 2.6 m/s; flight 85 s.
Plane 2: Wing area 0.042 m^2; cruise 2.1 m/s; flight 118 s.
Same: Mass, motor, and turns.

answer: Power required is drag times speed; flying slowly with low drag uses the motor's fixed energy at a lower rate. (4 points).

6. (4 points) Explain the typical shape of flight time versus launch torque.

Unit: Testing, the flight log, and competition day.
Worked-case data: Log data at a 9 m ceiling: torque 0.10 -> 95 s, 0.13 -> 118 s, 0.16 -> 104 s (ceiling bumps). Plan the first official flight.
Separate investigation observations:
Room A, 7 m ceiling: Best time at torque 0.10.
Room B, 11 m ceiling: Best time at torque 0.15.
Competition: Ceiling announced as 9 m.

answer: Time rises with torque until the plane reaches the ceiling, then falls because ceiling contact wastes energy. (4 points).

7. (4 points) How does propeller torque affect the flight?

Unit: Stability, trim, and circling flight.
Worked-case data: A plane stalls on every glide test. List adjustments in order to try.
Separate investigation observations:
Glide: Smooth and straight.
Powered launch: Plane rolls left and spirals down in the first 5 s.
After 5 s: Torque drops; plane recovers if still airborne.

answer: It rolls the plane opposite to propeller rotation, strongest during the burst, tightening or widening the circle. (4 points).

8. (4 points) Why back off from the maximum tested turns?

Unit: Rubber motors: storing and releasing energy.
Worked-case data: A team uses a 15:1 winder and tests show a motor breaks near 1,400 turns. How many cranks should they wind for a safe 8 percent back-off?
Separate investigation observations:
Motor A: 1.4 g, 30 cm long; 1,100 turns; climbed to ceiling, 78 s.
Motor B: 1.4 g, 38 cm long; 1,350 turns; stayed below ceiling, 104 s.
Ceiling: 8 m in both tests.

answer: Motors vary and weaken with use; winding near the breaking point risks a broken motor and damage. (4 points).

9. (4 points) When does timing of a flight end?

Unit: Rules, scoring, and the flight log bonus.
Worked-case data: A team with a complete flight log flies official flights of 96.4 s and 112.0 s. What is its score and tiebreaker?
Separate investigation observations:
Team A: Best flight 120 s, no flight log.
Team B: Best flight 105 s, complete flight log.
Team C: Best flight 112 s, partial flight log.

answer: When the plane touches the floor or when its lifting surfaces no longer support it, such as resting on a beam. (4 points).

10. (4 points) Why bring a second airplane?

Unit: Building light and strong.
Worked-case data: A built airframe weighs 7.4 g. The CG is slightly behind the target. How should the team meet the 8.0 g minimum?
Separate investigation observations:
Flight 1: Fresh build, 102 s.
Flight 6: After a week in a hot car, 61 s; plane turned sharply right.
Inspection: Right wing twisted 2 degrees.

answer: A crash or warp can end the event; a trimmed backup plane allows official flights within the 10-minute period. (4 points).

11. (6 points) Solve this case and explain your reasoning: A plane cruises at 1.2 m/s and its propeller effectively advances 0.24 m per turn. A motor has 1,200 usable turns. Estimate motor run time.

reasoning: Revolutions per second = 1.2 ÷ 0.24 = 5.0 rev/s.
Run time = 1,200 turns ÷ 5.0 rev/s = 240 s.
The burst spends turns faster, so real time is somewhat less. (3 points).

conclusion: About 240 s of motor run, shortened somewhat by the faster burst. (3 points).

12. (2 points) What best explains Plane 2's longer flight?

Supplied synthetic observations:
Plane 1: Wing area 0.030 m^2; cruise 2.6 m/s; flight 85 s.
Plane 2: Wing area 0.042 m^2; cruise 2.1 m/s; flight 118 s.
Same: Mass, motor, and turns.

A. It had a bigger motor

B. It weighed less

C. Lower wing loading let it fly slower with less power

Correct: C

13. (2 points) What launch torque is a sensible starting point?

Supplied synthetic observations:
Room A, 7 m ceiling: Best time at torque 0.10.
Room B, 11 m ceiling: Best time at torque 0.15.
Competition: Ceiling announced as 9 m.

A. 0.15

B. About 0.12-0.13

C. 0.10

Correct: B

14. (2 points) Define "Center of gravity (CG)" in the context of Wright Stuff.

answer: Point where the plane balances. (2 points).

15. (2 points) Define "Torque" in the context of Wright Stuff.

answer: Twisting force the motor applies to the propeller. (2 points).

16. (2 points) Define "Motor mass limit" in the context of Wright Stuff.

answer: Rubber and attachments at most 1.5 g. (2 points).

17. (2 points) Define "Motor stick" in the context of Wright Stuff.

answer: Beam carrying the motor tension. (2 points).

18. (2 points) Define "Slip" in the context of Wright Stuff.

answer: Difference between pitch and actual advance. (2 points).